

Accounting Research Center, Booth School of Business, University of Chicago

An Empirical Examination of Conference Calls as a Voluntary Disclosure Medium

Author(s): Richard Frankel, Marilyn Johnson and Douglas J. Skinner

Source: *Journal of Accounting Research*, Vol. 37, No. 1 (Spring, 1999), pp. 133-150

Published by: [Wiley](#) on behalf of [Accounting Research Center, Booth School of Business, University of Chicago](#)

Stable URL: <http://www.jstor.org/stable/2491400>

Accessed: 27-02-2016 10:22 UTC

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Accounting Research Center, Booth School of Business, University of Chicago and Wiley are collaborating with JSTOR to digitize, preserve and extend access to *Journal of Accounting Research*.

<http://www.jstor.org>

An Empirical Examination of Conference Calls as a Voluntary Disclosure Medium

RICHARD FRANKEL, MARILYN JOHNSON,
AND DOUGLAS J. SKINNER*

1. Introduction

Corporate conference calls are large-scale telephone conference calls during which managers make presentations to and answer questions from various market participants, usually about earnings. In this paper, we sample 1,056 corporate conference calls made by 808 firms during February–November 1995 to provide evidence on three questions: (1) whether conference calls provide information to stock market participants, (2) whether investors have equal access to the information provided during these calls, and (3) why managers of some firms hold conference calls while managers of other firms do not. We believe this research is important because managers' use of conference calls has grown enormously, yet we know little about how these calls affect investors.¹

*University of Michigan. We are grateful to First Call Corporation for providing data used in this study. Marilyn Johnson gratefully acknowledges the financial support of Ernst & Young, LLP. We thank Jeff Christensen (Christensen & Associates) and Stan Levine (First Call) for helpful discussions, and an anonymous referee, Lou Thompson (NIRI), Thomas Myers (TRW), and workshop participants at the American Accounting Association Meetings at Dallas, Emory, Michigan, Penn State, NYU, Stanford, and Washington (Seattle) for helpful comments. We also thank Joe Piotroski for expert research assistance.

¹Feldman [1996] reports that quarterly conference calls, typically after earnings releases, are routine, and the National Investor Relations Institute (NIRI [1996a]) states that conference calls are second only to press releases as a means of disseminating corporate information. The only other research on conference calls of which we are aware is a recent paper by Tasker [1997], whose work we discuss further below.

Whether conference calls provide information to market participants is basic to understanding their role in the corporate disclosure process. Although some argue that conference calls do not provide material information to investors beyond that in the accompanying press release (Feldman [1996] and Thompson [1997]), the fact that analysts and money managers participate in these calls suggests the opposite. Moreover, the issue of whether all investors have equal access to conference call information is moot if these calls are uninformative. We examine stock volatility and trading volume levels during conference calls to assess whether these calls are informative.

Commentators are divided on how conference calls affect investors' access to information. On one hand, some argue (Galant [1994], Feldman [1996], and Waroff [1994]) that conference calls, which involve many analysts and institutional investors and may even be open to anyone who wishes to participate, have improved the fairness of the information dissemination process since all call participants receive the same information at the same time. Others argue (Smith [1995], Gutner [1996], and Levitt [1998]) that because conference calls are limited to persons invited by the company to participate, they have improved the timeliness of disclosure to analysts and money managers while putting individual investors at a larger informational disadvantage. Analysts and money managers may transmit and/or immediately trade on information released during a conference call, while nonparticipating investors do not have this opportunity.² In addition, because transcripts of conference calls are rarely published, other investors may have to rely on analysts to process and disseminate the information. We assess whether trade size is unusually large during conference call periods, because large trades would likely indicate that some market participants (such as money managers and institutional investors) trade in real time on the basis of conference call information.

Our sample contains 1,056 conference calls, most (85%) of which are associated with earnings announcements. To investigate why managers of some firms choose to hold conference calls while others do not, we first describe firms that hold conference calls. We find that these firms are similar to firms that have more informative disclosure policies (e.g., see Lang and Lundholm [1993] and Lev and Penman [1990]); specifically, they are larger, have greater analyst following, and access capital markets more often than other firms. In addition, we find that firms in

² For an example in which this is alleged, see "Shut Out: Bank of New York Scrutinized Over Delay in Announcing Big Reserve," *Wall Street Journal* (June 21, 1996): C1. The Bank of New York announced a larger-than-expected increase in credit-card loss reserves during a conference call to analysts and institutional investors that occurred from 2 P.M. to 3 P.M. EDT on June 19, 1996. The bank issued a press release at 4:30 P.M. by which time its stock price had already fallen 2.5%. The price fell another 2.3% the next day. See also "Conference Calls to Big Investors Often Leave Little Guys Hung Up," *Wall Street Journal* (June 21, 1995): C1.

high-tech industries and with higher-than-average market-to-book ratios and sales growth rates are more likely to hold conference calls. This result suggests that firms with higher expected growth are more likely to use conference calls, perhaps because these firms have greater informational problems than other firms.

Tasker [1997] argues that firms with low “accounting quality” are more likely to hold conference calls, where accounting quality measures the extent to which financial statements provide information about a firm’s underlying operations. Tasker measures accounting quality using a composite measure based on market-to-book ratios, sales growth rates, and the extent to which book values and earnings explain security prices, and finds that firms with lower accounting quality are more likely to hold conference calls than other firms. Our evidence can be interpreted in a similar manner; overall, the evidence indicates that “growth” firms are more likely to hold conference calls, which may also reflect their lower “accounting quality.”

More important, we find that conference calls convey material information, as evidenced by unusually large return volatility and trading volume. In addition, we find that average trade size increases during conference calls, suggesting that larger investors trade in real time on the basis of information that is released during conference calls. This evidence implies that conference calls are informative but that not all investors have equal access to the information in these calls.

We discuss our hypotheses in the next section, our data and sample selection procedures in section 3, and in section 4 we compare the characteristics of firms that make conference calls with the characteristics of those that do not. Section 5 provides evidence on whether conference calls provide information to market participants and on whether investors have equal access to conference call information. We present our conclusions in section 6.

2. Hypothesis Development

2.1 WHY DO MANAGERS HOLD CONFERENCE CALLS?

Conference calls provide benefits to the company and its analysts. From the company’s perspective, conference calls save time and mitigate selective disclosure problems, because management can talk to hundreds of analysts and money managers simultaneously, and the investor relations staff receives fewer calls. From the analyst’s perspective, the benefits of conference calls include the opportunity to hear the questions of others (55%); the saving of time and money vis-à-vis traveling to meetings (45%); the receipt of timely information (13%); and the receipt of information at the same time as other investors (9%).³

³ Percentages taken from a survey of 122 analysts and portfolio managers by Christensen and Associates [1992].

Previous research finds that firms with more forthcoming disclosure policies are larger and more profitable than other firms (Lang and Lundholm [1993] and Lev and Penman [1990]) and that increased voluntary disclosure is associated with issuing securities to the public (Lang and Lundholm [1993], Ruland, Tung, and George [1990], and Frankel, McNichols, and Wilson [1995]). Since we view conference calls as a voluntary disclosure mechanism, we expect conference call firms to be larger, more profitable, and more likely to issue public capital.

Conference calls are often used to supplement mandated disclosures, in particular, quarterly earnings releases. It seems likely that conference calls will be used when such supplementation is most beneficial; for example, in helping analysts ascertain the extent to which earnings changes are permanent or transitory. Conference calls could thus be used to explain the implications of unusual or extraordinary items to analysts, and we predict that managers are more likely to use conference calls when earnings contain unusual or extraordinary items.

Managers of firms for which value is relatively more difficult for analysts to assess based on historical numbers may also be more likely to use conference calls. Specifically, managers of firms whose value is composed principally of intangible "growth options" (Myers [1977]) and/or managers of firms whose operations are growing rapidly may be more likely to use conference calls than managers of other firms. Because of their firms' greater informational problems, managers of these growth firms may view conference calls as a convenient medium for providing more information (including forward-looking information) to investors. These arguments are related to Lev's [1996] view that the value of high-tech firms (which tend to have higher growth rates and relatively more intangible assets) is less likely to be measured well by conventional accounting numbers, and to the arguments and evidence in Tasker [1997].

Why managers of these firms choose to hold conference calls rather than simply including the same information in press releases is less clear. One conjecture is that statements made during conference calls, being less formal than written press releases, are subject to a lower standard of legal liability than statements made during a press release. In addition, managers may use conference calls when they are unsure of the informational needs of investors (which seems more likely in newer, high-growth industries where the business environment is changing rapidly) and so prefer the flexibility of providing information in response to analysts' questions.

2.2 HOW DO CONFERENCE CALLS AFFECT STOCK MARKET PARTICIPANTS?

To disseminate corporate news, including earnings announcements, companies typically send press releases electronically to the newswires and to equity research databases such as *First Call* and *Bloomberg*, and follow up with "broadcast faxes" to hundreds of brokerage firms and institutions. These faxes often include the time and phone number of a

conference call, scheduled for a few hours after the press release. For routine events such as earnings announcements, a fax may also be sent in advance, to alert investors to the upcoming release and conference call. While previous research establishes that earnings releases are informative (e.g., Patell and Wolfson [1984]), it is not clear whether conference calls that follow these press releases provide *incremental* information to market participants, which is the question we address.

A typical conference call includes a 15–20-minute presentation by management, followed by a 30–45-minute question and answer session in which a moderator assigns questions to the management team. The sponsoring company usually restricts access (typically to analysts and money managers) and controls who asks questions.⁴

There are two reasons to expect that conference calls provide information to call participants. First, analysts' and institutional investors' continued participation indicates that these calls likely provide information incremental to that in the press release. Second, specific information such as detailed segment data, not in the press release, is sometimes disclosed in these calls. In addition, anecdotal evidence from the limited sample of calls we heard indicates that managers are more likely to make forward-looking statements during conference calls (often in response to analysts' questions) than to include them in the press release. In fact, many companies now begin conference calls with the cautionary language necessary to obtain safe harbor protection for forward-looking statements.⁵

Our tests of conference call information content examine whether stock prices and trading volume increase during the time these calls take place. A stronger test of conference call informativeness would assess whether the calls increase the overall level of information or whether the calls merely preempt information that is later released through other media. For example, if managers use conference calls to preempt the segment information in the 10K, there is no change in the total information available. However, if managers provide a more detailed segment breakdown in the conference call than subsequently provided in the 10K,⁶ the total amount of information available about the company increases as a result of the conference call. Since information revealed

⁴ The practice of screening questions is consistent with *NIRI's Standards and Guidance for Disclosure*, which points out that, "it is recognized that there will be some participants on the conference call who may ask irrelevant or misleading questions, who may not ask questions in good faith, or who may not want to ask a question at all but who simply may want to make a statement" (*NIRI* [1996a]).

⁵ These statements are based on several conference calls we listened to during 1996 and 1997.

⁶ The authors listened to a conference call in which company managers reported by-country, by-product segment data, much more detailed than the data subsequently provided in the company's annual report/10K.

during the course of a call becomes public through trading or dissemination by call participants, any increase in the total information set due to conference calls also represents an increase in the set of publicly available information.

To investigate this issue would require a measure of the total information available about firms, and tests of whether this information measure is affected by conference calls. Such tests, whether conducted in time series (examining whether the information measure increases after conference calls) or in cross-section (comparing the amount of information available about conference call and nonconference call firms), are fraught with selection and endogeneity problems. Our tests do not distinguish whether conference calls increase the total amount of information available about firms or merely change the ordering and/or timeliness with which various market participants receive a given amount of information.

Conference calls may place individual investors at an informational disadvantage. A survey by *NIRI* (*NIRI* [1996b]) indicates that, of a sample of 147 companies that use conference calls, 97% invite buy-side and/or sell-side analysts, 92% (77%) invite current (potential) institutional holders, 38% invite brokers, 10% invite the media, and 22% invite current individual stockholders. To test whether conference calls allow equal access to all investors, we examine trade size during conference call periods, expecting that average trade size increases during conference calls if institutional investors use the information they receive to trade in real time.⁷

3. Data and Sample Description

Our sample contains conference calls from February to November 1995 listed in *First Call*, a real-time equity research database marketed by First Call Corporation.⁸ *First Call* screens brokerage research notes, company announcements, and other sources to prepare its daily listing of conference calls. During February–November 1995, *First Call* identified 1,440 conference call announcements, which we matched with a header file constructed from *CRSP* data to obtain *CUSIPs*. Seventy-one observations did not match and 313 *CUSIPs* were not available on *Compustat*. Our tests are based on subsets of the remaining 1,056 (808) observations (firms) for which the required data are available.

We classify the information discussed in the conference call based on the information in the accompanying press release (obtained from *Bloomberg*) and find that most (85%) of the sample conference calls are

⁷ One money manager who practices “earnings momentum” strategies claims that his firm routinely trades in real time on information released during conference calls.

⁸ *First Call* provides on-line, instantaneous access to Wall Street research and corporate releases. It is a joint venture between Thomson Corp. and nine leading brokerage firms.

associated with earnings releases. The other releases are primarily announcements of corporate control events. Our tests focus on the calls associated with earnings releases; results are similar when we consider all conference calls.

To provide evidence on why some managers hold conference calls while others do not, we compare our sample firms to *Compustat* firms that do not appear in *First Call*, which we label “nonconference call” firms. We randomly selected 224 of the 12,228 nonconference call firms and telephoned them to learn their disclosure practices. Since we also use this subsample as a control group to examine intraday price, volume, and trade size movements following quarterly earnings announcements, we called only those firms with at least one quarterly earnings announcement time stamp available from *Bloomberg* during March–June 1995. We were unable to locate an earnings announcement or telephone number for 33 firms, and 3 firms refused to take our call.

Of the remaining 188 firms, 172 (91.7%) confirmed they did not hold conference calls during 1995, while 16 firms (8.3%) indicated they did. Of these 16 firms, 7 began holding conference calls during 1995 but could not identify when. Of the remaining 9 firms, none provided a reason why its call was not listed on *First Call*. Overall, the high proportion of firms (91.7%) confirming they did not hold conference calls in 1995 gives us confidence that our conference call vs. nonconference call classification is reliable.⁹ Personnel from nonconference call firms who were willing to explain their company’s decision stated that sell-side analyst coverage was too small to warrant conference calls, that they had no complicated information to convey, that conference calls are too expensive, and that portfolio managers and sell-side analysts prefer one-on-one calls.

4. Characteristics of Firms Making Conference Calls

To investigate why managers of some firms hold conference calls while others do not, we compare conference call and nonconference call firms in table 1. Consistent with previous voluntary disclosure research, conference call firms are larger and more profitable than other firms. The average (median) total assets of conference call firms is \$6,031 million (\$560 million) compared to \$1,696 million (\$94 million) for nonconference call firms. Return on assets (*ROA*) is also higher for conference call firms: average (median) *ROA* is 5% (4%) for conference call firms compared to –9% (2%) for nonconference call firms. Twelve percent of conference call firms report losses in fiscal 1994 compared to 29% of

⁹ To the extent that there is misclassification, it is likely that conference call firms are incorrectly classified as nonconference call firms, reducing the power of our tests.

TABLE 1
Summary Statistics for Conference Call and Nonconference Call Firms

Variable	Firm Type		Z-Statistic for Wilcoxon Rank-Sum Test	T-Test for Difference of Means
	Conference Call Firms	Nonconference Call Firms		
<i>Total Assets</i>				
Median	560.34	94.05	20.33***	9.22***
Mean	6,030.98	1,696.20		
Obs.	(726)	(5,886)		
<i>ROA</i>				
Median	0.04	0.02	12.98***	2.94***
Mean	0.05	-0.09		
% < 0	12.11%	29.29%		
Obs.	(718)	(5,569)		
<i>Sales Growth</i>				
Median	0.09	0.06	6.16***	2.07**
Mean	0.16	0.12		
Obs.	(458)	(3,080)		
<i>Price-Earnings</i>				
Median	13.94	12.22	4.085***	0.223
Mean	31.92	51.52		
Obs.	(528)	(2,610)		
<i>Market-to-Book</i>				
Median	1.82	1.28	10.79***	0.00
Mean	2.62	3.56		
Obs.	(868)	(5,549)		
<i>D/E</i>				
Median	1.15	1.24	-1.46	-1.60
Mean	2.49	4.57		
Obs.	(670)	(4,835)		
<i>StdROA</i>				
Median	0.03	0.05	-6.38***	-1.38
Mean	0.06	0.35		
Obs.	(467)	(3,457)		
<i>INTCOV</i>				
Median	4.97	2.02	9.86***	0.39
Mean	44.13	26.53		
Obs.	(618)	(3,786)		
<i>SPI/TA</i>				
Median	0	0	-1.69*	1.08
Mean	-0.01	-0.02		
% < 0	27.56%	22.33%		
% 0	38.25%	32.48%		
Obs.	(664)	(5,046)		
<i>XI/TA</i>				
Median	0	0	-4.25**	-0.56
Mean	0.03	0.06		
% < 0	11.82%	8.85%		
% 0	21.31%	23.63%		
Obs.	(685)	(5,162)		
<i>Common Stock</i>				
Median	0	0	13.87***	13.97***
Mean	0.16	0.04		
Obs.	(808)	(12,228)		

TABLE 1—continued

<i>Debt</i>				
Median	0	0	25.08***	25.70***
Mean	0.24	0.04		
Obs.	(808)	(12,228)		
<i>Other</i>				
Median	0	0	13.91***	14.01***
Mean	0.07	0.01		
Obs.	(808)	(12,228)		
<i>Number of Analysts</i>				
Median	8	3	20.766***	18.16***
Mean	10.97	5.64		
Obs.	(894)	(3,027)		

Table values represent means, medians, and number of observations of various variables for conference call and nonconference call sample firms. The number of observations is shown in parentheses. Conference call firms appeared in the *First Call* conference call database between February and November 1995 and have the required *Compustat* data. Nonconference call firms are firms with the required *Compustat* data that did not appear in the *First Call* conference call database. *Total Assets* are measured at the end of fiscal 1994. *ROA* is the return on total assets for fiscal 1994. *Sales Growth* is the geometric mean of annual sales growth between fiscal years 1990 and 1994. *StdROA* is the variance in annual return on assets between fiscal 1990 and 1994. *INTCOV* is the interest coverage ratio for fiscal 1994 computed by dividing cash flow from operations by interest paid. *SPI/TA* is the amount of special items for fiscal 1994 divided by total fiscal 1994 assets. *XI/TA* is the amount of extraordinary items for fiscal 1994 divided by total fiscal 1994 assets. *Common Stock* is an indicator variable taking a value of one if the company had a common stock offering in the Securities Data Corporation (*SDC*) database, and zero otherwise. *Debt* is an indicator variable taking a value of one if the company had a debt offering in the *SDC* database. *Other* is an indicator variable taking a value of one if the company had a preferred stock offering, convertible security offering, or other non-public-common-stock or non-straight-debt offering. The *SDC* database covers the period from January 1, 1995 to May 15, 1996. *Market-to-Book* is computed using shareholders' equity and prices at the beginning of fiscal 1995.

***, **, * signify that the conference call and nonconference call distributions of the given variable are significantly different at the 1%, 5%, and 10% levels, respectively (two-tailed).

nonconference call firms. All of these differences are statistically significant at the 5% level or better.

Conference call firms exhibit relatively rapid growth: average (median) annualized sales growth is 16% (9%) for conference call firms compared to 12% (6%) for nonconference call firms, differences significant at the 1% level. In addition, we find that median price-earnings and market-to-book ratios are reliably higher for conference call firms at the 1% level or better, although the means are not (see also Tasker [1997]). These differences suggest that market participants expect higher future growth rates for conference call firms, that their managers face greater informational problems as a result, and so provides a possible reason managers of these firms are more likely to hold conference calls. Alternatively, analysts may simply demand more information (and perhaps more forward-looking information) when firms are growing more rapidly.

We measure the risk characteristics of conference call firms using the debt-equity ratio, the standard deviation of *ROA*, and a cash-based interest coverage ratio. Both samples have similar debt/equity ratios; however, conference call firms are less risky than nonconference call firms on the other two measures. The average (median) standard deviation of *ROA* is 6% (3%) for conference call firms and 35% (5%) for nonconference call firms; the difference in medians (means) is significant at the

1% (17%) level. Similarly, interest coverage is higher for conference call firms, although only the difference in medians is statistically significant at the 5% level: average (median) interest coverage is 44.1 (5.0) for conference call firms compared to 26.5 (2.0) for nonconference call firms.

While most sample firms report no special or extraordinary items, conference call firms are more likely (28% vs. 22%) to report *negative* special items. The Z-statistic for the Wilcoxon two-sample test is -1.69, significant at the 10% level (two-tailed). Similarly, 12% of conference call firms report negative extraordinary items compared to 9% for nonconference call firms (Wilcoxon $Z = -4.25$). These comparisons suggest that when earnings contain unusual or extraordinary components, analysts demand information to parse out the implications of these factors for their estimates of future earnings, and managers use conference calls to supply this information.¹⁰ This evidence is consistent with the more general idea that analysts try to ascertain the extent to which current earnings components are permanent (part of “core” earnings) or transitory, and look to management to provide them with guidance on this issue, often during conference calls.

Conference call firms go to the capital markets more often than nonconference call firms. From the beginning of 1995 through the middle of 1996, 16% of conference call firms raised capital (vs. 4% of nonconference call firms), 24% issued public debt (vs. 4%), and 7% issued some other type of security (vs. 1%); all of these differences are significant at the 1% level. This result is consistent with prior research, which suggests that firms issuing securities increase their disclosure levels.

As expected given that conference call firms are larger and more profitable, analyst following is also larger for these firms—the mean (median) number of analysts following conference call firms is 11 (8), while for nonconference call firms analyst following is 6 (3) analysts. Assuming these results are not due solely to these firms’ size and profitability, they are again consistent with results from the voluntary disclosure literature, which suggest that firms with more forthcoming disclosures tend to have higher analyst following (Lang and Lundholm [1996]), although the direction of causality cannot be determined from our evidence.

Probit models of the decision to hold conference calls are reported in table 2. We report results with and without the market-to-book variable since the results are sensitive to its inclusion. Results of the first estimation indicate that conference call firms tend to be larger, with higher market-to-book ratios and greater analyst following, than nonconference call firms. The only other variable significant in this estimation is the debt issuance variable; this result is somewhat surprising since we expected that equity issuances would be more likely to affect disclosure.

¹⁰ It may be less costly for managers to do this in response to analysts’ questions during a conference call than through a press release (where they would have to anticipate analysts’ questions).

TABLE 2
Probit Estimation of the Relation between Conference Calls and Firm Characteristics

Variable	Specification 1		Specification 2	
	Estimate	T-Statistic	Estimate	T-Statistic
Intercept	-1.902	-7.986***	-1.575	-7.847***
Information Environment				
<i>LNASSET</i>	0.114	2.602**	0.037	1.63*
<i>MtB</i>	0.063	2.624***		
<i>NANALYST</i>	0.036	4.518***	0.043	6.129***
Profitability Indicators				
<i>ROA</i>	0.432	0.829	0.729	1.429*
<i>SGROWTH</i>	0.009	0.693	0.009	0.673
Riskiness Indicators				
<i>StdROA</i>	0.287	0.690	0.568	1.410*
<i>D/E</i>	-0.020	-1.135	-0.004	-0.479
<i>INTCOV</i>	0.000	-0.826	0.000	-0.797
One-Time Events				
<i>XI/TA</i>	-0.391	-0.634	-0.627	0.624
<i>SPI/TA</i>	-0.674	-0.585	-1.109	-0.984
External Financing Variables				
<i>PUCS</i>	0.093	0.693	0.136	1.025
<i>DEBT</i>	0.217	1.853**	0.225	1.923**
<i>OTHER</i>	-0.084	-0.679	-0.085	-0.683
Observations = 1,251				
Maddala R^2	0.125		0.118	
Likelihood Ratio χ^2	166.376		156.394	

This table presents the results of a probit estimation of the equation below with and without market-to-book. *CCALL* is coded one of the sample firm had a conference call in the *First Call* conference call database between February and November 1995, and zero otherwise. *LNASSET* is the log of total assets at the end of fiscal 1994. *MtB* is the market-to-book ratio computed as price divided by shareholders' equity. *NANALYST* is the number of analysts following the company in the January 1995 statistical period on the *I/B/E/S* consensus tape. *ROA* is the return on total assets for fiscal 1994. *SGROWTH* is the geometric mean of annual sales growth between fiscal years 1991 and 1994. *StdROA* is the variance in annual return on assets between fiscal 1990 and 1994. *D/E* is the debt-to-equity ratio at the end of fiscal 1994. *INTCOV* is the interest coverage ratio for fiscal 1994 computed by dividing cash flow from operations by interest paid. *XI/TA* is the amount of extraordinary items for fiscal 1994 divided by total fiscal 1994 assets. *SPI/TA* is the amount of special items for fiscal 1994 divided by total fiscal 1994 assets. *PUCS* is an indicator variable taking a value of one if the company had a public common stock issuance transaction in the Securities Data Corporation (*SDC*) database, and zero otherwise. *DEBT* is an indicator variable taking a value of one if the company had a debt issuance transaction in the *SDC* database, and zero otherwise. *OTHER* is an indicator variable taking a value of one if the company had a non-common-stock and non-debt-issuance transaction in the *SDC* database, and zero otherwise. The portion of the *SDC* database covering the period between January 1995 and May 1996 is used in this sample.

***, **, * signify that the coefficients on the variables are significantly different from zero at the 1%, 5%, and 10% levels, respectively (one-tailed).

The R^2 shown below is the Maddala R^2 , calculated as $1 - (L_\omega/L_\Omega)^{2/n}$, where L_ω is the likelihood function maximized with respect to all the parameters and L_Ω is the function maximized only with respect to the intercept. The Likelihood Ratio χ^2 statistics are distributed χ^2 with 12 degrees of freedom (11 for the estimation excluding *MtB*), and each have a probability value less than 0.0001.

$$\text{pr}(\text{CCALL}) = \beta_0 + \beta_1 \text{LNASSET} + (\beta_2 \text{MtB}) + \beta_3 \text{NANALYST} + \beta_4 \text{ROA} + \beta_5 \text{SGROWTH} + \beta_6 \text{StdROA} + \beta_7 \text{D/E} + \beta_8 \text{INTCOV} + \beta_9 \text{XI/TA} + \beta_{10} \text{SPI/TA} + \beta_{11} \text{PUCS} + \beta_{12} \text{DEBT} + \beta_{13} \text{OTHER} + \varepsilon.$$

When we exclude the market-to-book variable, two additional variables become significant at the 10% level—both average *ROA* and its standard deviation are higher for conference call firms.

Industry comparisons using three-digit *SIC* codes (results not tabulated) indicate that managers of firms in the steel, “high-tech” (computer

hardware and software, communications, and electronics), and insurance industries are more likely to hold conference calls than managers of other firms. Firms in the oil and gas and banking industries are less likely to hold conference calls than other firms. The explanation for these patterns is not obvious, although we interpret the predominance of conference calls in high-tech industries as supporting the idea that these industries are more difficult for analysts to understand using historical financial data because of their high rate of change, level of technical difficulty, and/or the intangible nature of their assets.

5. *Intraday Market Responses to Press Releases and Conference Calls*

This section provides evidence on whether conference calls provide information incremental to that in the accompanying press release and whether investors have equal access to conference call information. Such evidence is important in assessing whether conference calls give some market participants a temporary informational advantage.

We use measures of return volatility and trading volume to assess the information content of conference calls. Holthausen and Verrecchia [1990] demonstrate that unexpected return variance and trading volume are equally valid measures of information content (defined as a signal that alters investors' beliefs); that is, either metric can be used to assess information content. Following research on intraday data (Lee, Ready, and Seguin [1994]), we report three measures of volatility in table 3. The first is the absolute value of the return during the conference call window. The second (*HILO*) is the absolute difference between the highest and lowest trade price during the conference call window, which some argue is a more efficient estimator of volatility (Wiggins [1991]). Third is the number of revisions to the midspread (the midpoint between the bid and ask price) of the *NYSE* specialist's quote (*SPDREV*) which measures the frequency of price changes during the conference call window. Trading volume (also reported in table 3) is measured as total trading volume during the conference call window.

We compare the distributions of these variables during the conference call and a control period. Since we cannot observe the duration of each conference call, we define our conference call window as beginning 15 minutes before the call start time and ending 60 minutes after the start. Most calls start with a 15–20-minute presentation, followed by a question and answer period that typically lasts another 30–45 minutes. The control period is the 75 minutes that begins 90 minutes before the time of the conference call and ends 15 minutes before the time of the call. Thus, our control period lies between the time of the press release and the time of the conference call.¹¹ This benchmark addresses the

¹¹ According to the Christensen and Associates [1992] survey, most companies schedule conference calls at least 90 minutes after they plan to issue their releases electronically.

TABLE 3
Intraday Market Volatility Measures and Trading Volume During Announcement, Conference Call, and Control Periods

Panel A: Earnings Announcement Sample				
	Reference Period	Announcement Window ($t_a - 15, t_a + 60$)	Control Period ($t_c - 90, t_c + 15$)	Conference Call Window ($t_c - 15, t_c + 60$)
<i>Returns</i>				Minus Control Period
Mean	0.004361	0.017698	0.007765	0.012987
Median	0.001865	0.01064	0.004339	0.005378
Observations	164	164	164	164
<i>HIL0</i>				
Mean	21.342¢	64.215¢	37.005¢	42.550¢
Median	6.250¢	50.000¢	25.000¢	25.000¢
Observations	164	164	164	164
<i>SPDREV</i>				
Mean		12.103	6.348	8.030
Median		9	4	5
Observations		164	164	164
<i>Volume</i>				
Mean	36.838	195,470	91,231	113,481
Median	10,675	52,600	25,750	36,050
Observations	164	164	164	164
Panel B: Pseudo-Conference-Call Sample				
	Announcement Window ($t_a - 15, t_a + 60$)	Control Period ($t_a + 145, t_a + 220$)	Pseudo-Conference-Call Window ($t_a + 220, t_a + 295$)	Pseudo-Conference-Call Minus Control Period
<i>Returns</i>				
Mean	0.0175	0.0128	0.0122	-0.0005 (-0.239)
Median	0.0095	0.0028	0	0 (0.6678)
Observations	178	178	178	178
<i>Volume</i>				
Mean	22,259	19,735	18,792	-938.33 (-0.267)
Median	3,900	2,500	2,800	0 (0.773)
Observations	178	178	178	178

Panel A of this table presents volatility measures and volume during announcement, conference call, and selected control periods. $|Returns|$ is the absolute value of returns. *HIL0* is the absolute difference between the highest and lowest trade price in cents. *SPDREV* is the number of revisions in the average of the bid and the ask. *Volume* is trading volume. Announcements chosen for inclusion are those closest to the time of the included conference call. Announcements are obtained from the *Bloomberg* database. Intraday volume and return data are obtained from the *NYSE Trade and Quote (TAQ)* database. Conference calls include the conference calls of all firms in the *First Call* conference call database between February and November 1995 that have required *TAQ* and *Bloomberg* data as well as nonoverlapping announcement windows, control periods, and conference call windows. t_a is the time of the announcement. t_c is the scheduled time of the conference call. The numbers added and subtracted from the times are in minutes, e.g., the announcement window begins 15 minutes before the announcement and ends 60 minutes after the announcement. The reference period is composed of the measurement windows four days to one day before the conference call day corresponding to the same time of day as the conference call window. Numbers shown under reference period are the averages of the measurement windows over these four days. Numbers shown in parentheses next to means are two-tailed p -values from t -tests. Numbers shown in parentheses next to medians are two-tailed p -values from sign-rank tests.

Panel B of this table presents a comparison of $|Returns|$ and *Volume* during two intervals for a sample of earnings announcements that are not followed by conference calls. The 75-minute pseudo-conference-call period begins 220 minutes after an earnings announcement. The 75-minute control period begins 145 minutes after the earnings announcement.

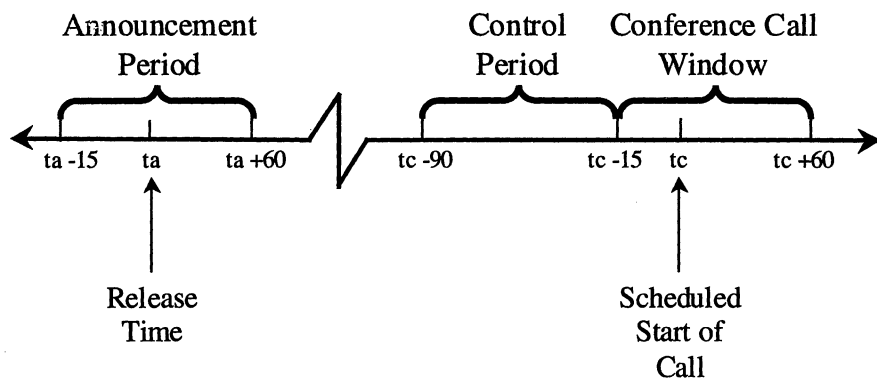


FIG. 1.—Time line for announcement, control, and conference call windows (time is measured in minutes).

concern that we would observe unusually high variance and volume during the conference call period simply because it takes a day or two for volume and variance to return to “normal” levels after an earnings release. For comparison purposes, we also report variance and volume during the announcement period, defined as the period from 15 minutes before the press release through 60 minutes afterward. All three periods are nonoverlapping (see figure 1). Additionally, we report variance and volume statistics computed during the four days immediately preceding the announcement date, computed at the same time of day as the conference call.¹²

The results of the volatility tests, presented in table 3, indicate that return volatility is higher during the conference call period than during the control period. Volatility follows a *U*-shaped pattern; it is highest at the time of the earnings release, declines during the control period, and increases again during the conference call window. The mean (median) absolute return during the conference call is 1.23% (.54%) compared to .78% (.43%) during the control period, differences significant at the 1% and 8% levels, respectively. (We report two-tailed *p*-values throughout.) The numbers for the earnings release window are 1.77% (1.06%). Results are similar for *HILO* but not as strong; the mean (median) difference is significant at the 16% (3%) level. Results are strongest for the spread revisions: the mean (median) spread revision is 8.0 (5) during the conference call window compared to 6.3 (4) during the control period, differences significant at the 1% level. Overall, table 3 demonstrates that prices are unusually volatile during conference calls, which indicates that these calls are informative.¹³ Return volatility during conference

¹² Previous research indicates that there are important “time of day” effects in intraday price and volume metrics (e.g., see Jain and Joh [1988] and Wood, McInish, and Ord [1985]).

¹³ These tests are conservative; return variance declines slowly after announcements, so even if conference calls are not informative, we would expect return variance during the

calls tends to be smaller than that during the earnings announcement window, indicating that the original press release is more informative.

The trading volume results are also reported in table 3 and also indicate that conference calls are informative. Both mean and median trading volumes are noticeably higher during the conference call window than during either benchmark period. The mean difference (vs. the control period) is significant at the 7–8% level and the difference in medians is significant at the 1% level in two-tailed tests. These differences are also economically significant. Mean (median) volume during the conference call window of 113 (36) thousand shares is 24% (38%) higher than mean (median) control period volume of 91 (26) thousand shares. Trading volume during the conference call period is three to four times higher than that during the reference period (which measures “normal” volume levels). Overall, table 3 shows that conference calls convey information to market participants beyond the information provided by the press release, although it does not tell us whether investors have equal access to this information.

To address the equal access question, table 4 presents evidence on average trade size which indicates that average trade size is larger during conference calls than during control periods. Mean (median) trade size is 2,540 (1,674) shares during the conference call compared to 2,035 (1,020) shares during the control period. These differences are statistically significant at the 5% and 1% levels, respectively. Trade size is also substantially larger during conference calls than during “normal” periods: mean (median) trade size is 2,540 (1,674) shares during the conference call and 1,478 (1,204) shares during “normal” periods. These results are consistent with the view that large investors trade in real time on information released during conference calls.

To ensure that these results are attributable to conference calls per se and do not reflect normal trading patterns after earnings releases, we replicate our analyses on the sample of nonconference call firms. Specifically, for the 172 nonconference call firms we assign a “pseudo-conference-call period” as the 75 minutes that begin 220 minutes after an earnings announcement (220 minutes is the average delay between an earnings announcement and a conference call for our sample of conference call firms).

Results for the nonconference call firms, reported in panel B of both table 3 and table 4, indicate that the volatility, volume, and trade size measures are not significantly different (at the 10% level) for the pseudo-conference-call and control periods. These results provide assurance that our results are attributable to conference calls per se.

control period to be higher than during the conference call simply because the control period is closer to the time of the press release.

TABLE 4
Intraday Market Trade Size Measures During Announcement, Conference Call, and Control Periods

Panel A: Earnings Announcement Sample				
	Reference Period	Announcement Window ($t_a - 15, t_a + 60$)	Control Period ($t_c - 90, t_c - 15$)	Conference Call Window ($t_c - 15, t_c + 60$)
Trade Size				Conference Call Minus Control Period
Mean	1,478	2,096	2,035	2,540
Median	1,204	1,600	1,020	1,674
Observations	164	164	164	164
Panel B: Pseudo-Conference-Call Sample				
		Announcement Window ($t_a - 15, t_a + 60$)	Control Period ($t_a + 145, t_c + 220$)	Pseudo-Conference-Call Window ($t_a + 220, t_a + 295$)
Trade Size				Pseudo-Conference-Call Minus Control Period
Mean		1,389	1206.5	1422.6
Median		825	530	600
Observations		178	178	178

Panel A of this table presents trade size statistics during announcement, conference call, and control periods. Announcements chosen for inclusion are those closest to the time of the included conference call. Announcements are obtained from the *Bloomberg* database. Trade size data are obtained from the *NYSE Trade and Quote* (TAQ) database. The trade size for an observation is the total volume during the measurement window divided by the total number of trades.

Panel B presents a comparison of trade size during two intervals for a sample of earnings announcements that are not followed by conference calls. The 75-minute pseudo-conference-call period begins 220 minutes after an earnings announcement. The 75-minute control period begins 145 minutes after the earnings announcement.

We also compare the magnitude of the stock price reaction to earnings releases for the two sets of firms. If managers use conference calls only after large earnings surprises, we might expect that earnings releases accompanied by conference calls have greater information content than other earnings releases. There is little evidence that this is the case—the mean (median) absolute value of the announcement window return for the conference call firms is 1.77% (1.06%), very similar to the numbers of 1.75% (.95%) for the nonconference call firms. The volume and trade size numbers tend to be larger for the conference call firms; this is expected because these firms are larger than nonconference call firms.

Finally, we compute average returns during the various periods as well as correlations between the average returns in the announcement and conference call windows (not reported in the tables). These tests explore whether managers consistently place a positive “spin” on results during conference calls (in which case average returns are expected to be positive) or whether there is any systematic relation between information in the press release and that provided during the conference call (e.g., if the conference call were used to counter a perceived overreaction to the press release, the correlation would be negative). Average returns during all periods are close to zero. We find that the average (median) return during the earnings announcement window is $-.05\%$ (0%), while during the conference call window it is $.13\%$ (0%); differences between these returns are not statistically significant at the 5% level. There is no evidence of any significant correlation between returns during the press release and conference call periods.

6. Conclusion

This paper provides evidence on the characteristics of firms that hold conference calls, on whether these calls are informative, and on whether these calls provide equal access to all investors. Firms that hold conference calls tend to be relatively larger, more profitable, and more heavily followed by analysts; they also access the capital markets more often than other firms. These results are broadly consistent with those of previous voluntary disclosure research. In addition, and consistent with Tasker [1997], we find that conference call firms are more likely to be in high-tech industries, to grow more rapidly, and to have higher market-to-book ratios than other firms. To the extent that these variables proxy for informational asymmetries, we interpret these results as suggesting that conference calls help managers reduce such asymmetries.

We find that conference calls provide information to market participants over and above the information contained in the corresponding press release, as evidenced by elevated return variances and trading volumes. Additionally, average trade size is higher during the time of conference calls, suggesting that larger investors trade on the released information. These results indicate that material information is released

during conference calls and that a subset of large investors trade on this information in real time. Unless all stockholders are provided equal access to these calls, this conduct seems inconsistent with the SEC's espoused goal of providing a "level playing field" with regard to corporate disclosures.

REFERENCES

- CHRISTENSEN AND ASSOCIATES. *Survey on Teleconferencing*. Stamford, Conn.: Christensen and Associates, 1992.
- FELDMAN, B. "It's Your Call." *Investor Relations* (December 1996): 35-37.
- FRANKEL, R.; M. McNICHOLS; AND G. P. WILSON. "Discretionary Disclosure and External Financing." *The Accounting Review* (January 1995): 135-50.
- GALANT, D. "The Technology Trap." *Institutional Investor* (May 1994): 141-45.
- GUTNER, T. "How to Keep the Little Guy in the Loop." *Business Week* (July 29, 1996): 32.
- HOLTHAUSEN, R. W., AND R. E. VERRECCHIA. "The Effect of Informedness and Consensus on Price and Volume Behavior." *The Accounting Review* (January 1990): 191-208.
- JAIN, P., AND G. JOH. "The Dependence Between Hourly Prices and Trading Volume." *Journal of Financial and Quantitative Analysis* (September 1988): 269-84.
- LANG, M., AND R. LUNDHOLM. "Cross-Sectional Determinants of Analyst Ratings of Corporate Disclosures." *Journal of Accounting Research* (Autumn 1993): 246-71.
- _____. "Corporate Disclosure Policy and Analyst Behavior." *The Accounting Review* (October 1996): 467-92.
- LEE, C. M. C.; M. J. READY; AND P. J. SEGUIN. "Volume, Volatility, and New York Stock Exchange Trading Halts." *Journal of Finance* (March 1994): 183-214.
- LEV, B. "The Boundaries of Financial Reporting and How to Extend Them." Working paper, New York University, August 1996.
- LEV, B., AND S. H. PENMAN. "Voluntary Forecast Disclosure, Nondisclosure, and Stock Prices." *Journal of Accounting Research* (Spring 1990): 49-76.
- LEVITT, A. "A Question of Integrity: Promoting Investor Confidence by Fighting Insider Trading." Prepared remarks to the S.E.C. Speaks Conference, February 27, 1998.
- MYERS, S. "Determinants of Corporate Borrowing." *Journal of Financial Economics* (November 1977): 147-75.
- NATIONAL INVESTOR RELATIONS INSTITUTE. *Standards and Guidance for Disclosure*. Vienna, Va.: NIRI, 1996a.
- _____. *Technology Survey*. Vienna, Va.: NIRI, 1996b.
- PATELL, J. M., AND M. A. WOLFSON. "The Intraday Speed of Adjustment of Stock Prices to Earnings and Dividend Announcements." *Journal of Financial Economics* (June 1984): 223-52.
- RULAND, W.; S. TUNG; AND N. E. GEORGE. "Factors Associated with the Disclosure of Managers' Forecasts." *The Accounting Review* (July 1990): 710-21.
- SMITH, R. "Conference Calls to Big Investors Often Leave Little Guys Hung Up." *Wall Street Journal* (June 21, 1995): C1.
- TASKER, S. "Voluntary Disclosure as a Response to Low Accounting Quality: Evidence from Quarterly Conference Call Usage." Working paper, Cornell University, February 1997.
- THOMPSON, L. "President's Report." *National Investor Relations Institute Update* (July 1997): 2.
- WAROFF, D. "The Well-Connected IR Officer." *Institutional Investor* (May 1994): 134-38.
- WIGGINS, J. "Empirical Tests of the Bias and Efficiency of the Extreme-Value Variance Estimator for Common Stocks." *Journal of Business* (July 1991): 417-32.
- WOOD, R. A.; T. H. MCINISH; AND J. K. ORD. "An Investigation of Transactions Data for NYSE Stocks." *Journal of Finance* (July 1985): 723-39.